TEXTBOOK · LEVEL 5

OHS Instrumentation & A.I.A. Practicals

Modular — 8 modules · 32 credits · Entry: Level 5 or hygiene technician experience

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NQF LEVEL 6

Diploma in Applied OHS Instrumentation, Hygiene Telemetry & Approved Inspection Authority (A.I.A.) Practicals

Anchors: GN 156 of 1997 (A.I.A. monitoring); Atomic Energy and Radiation Protection Act 5 of 2005 (NRPA).

Formal vocational delivery and pre-accreditation submission: programmes commence only after NTA provider registration has been granted; NQA accreditation is pursued in parallel and will be announced on the public register. This pack is a pre-accreditation curriculum draft built on the Oryx template system.

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Programme outcomes

On completion of this level the learner can plan, execute and supervise the module outcomes below under normal workplace supervision, and produce the associated records to audit standard.

MODULES

Quantitative Exposure Math — TWA, STEL, Action Levels & Excursion Limits

NTA reference: GN 156

8-hr/12-hr time-weighted averages, mixture corrections, excursion management.

Exposure math is the hygiene licence: 8-hour and 12-hour TWAs, STEL comparisons, mixture corrections, action levels and excursion limits calculated by hand and defended line by line.

Practical video recording scheduled — the online reader reserves a demonstration slot for this module.

Multi-Gas Sensor Telemetry — O2, LEL, CO, H2S, SO2

NTA reference: GN 156

Catalytic bead, infrared and PID principles; calibration, cross-sensitivity, bump testing.

Sensor telemetry starts at the physics: catalytic bead, infrared and PID sensing principles, calibration gas discipline, cross-sensitivity traps and the bump-test regime that keeps readings honest.

Practical video recording scheduled — the online reader reserves a demonstration slot for this module.

Gravimetric Air Sampling & Optical Dust Telemetry

NTA reference: GN 156

Cyclone samplers, flow calibration, respirable crystalline silica and uranium dust.

Gravimetric sampling runs on flow: cyclone selection, rotameter calibration, filter handling and the respirable crystalline silica and uranium dust analyses that Namibian mines require.

Practical video recording scheduled — the online reader reserves a demonstration slot for this module.

Acoustic Dosimetry, Octave-Band Analysis & 85 dBA Noise Zoning

NTA reference: GN 156

Dosimetry practice, octave bands, zoning and hearing conservation.

Noise work is dosimetry plus interpretation: full-shift dosimetry, octave-band analysis for engineering decisions, and the 85 dBA zoning that drives hearing conservation under GN 156.

Practical video recording scheduled — the online reader reserves a demonstration slot for this module.

Thermal Stress Instrumentation — WBGT Arrays, Anemometry & Lux Grid Surveys

NTA reference: GN 156

WBGT metering, work-rest regimes, ventilation velocity and illumination surveys.

Thermal and visual environments are measured, not guessed: WBGT arrays with work-rest regimes, anemometry for duct velocity, and lux grid surveys against task lighting requirements.

Practical video recording scheduled — the online reader reserves a demonstration slot for this module.

Whole-Body & Hand-Arm Vibration (WBV/HAV) & Ergonomic Biomechanics

NTA reference: GN 156

Accelerometry, exposure points, biomechanical assessment.

Vibration assessment combines accelerometry with biomechanics: whole-body and hand-arm exposure points, tool selection, and the ergonomic reasoning that turns data into work redesign.

Practical video recording scheduled — the online reader reserves a demonstration slot for this module.

Radiation Protection Telemetry — GM Survey Meters, TLD Dosimeters & NORM + IoT PDS

NTA reference: NRPA (Act 5 of 2005)

Survey meters, dosimetry, NORM management, proximity detection systems.

Radiation telemetry under the NRPA: Geiger-Müller survey practice, TLD dosimetry programmes, NORM management in mining streams, and IoT proximity detection as an engineering control layer.

Practical video recording scheduled — the online reader reserves a demonstration slot for this module.

A.I.A. Metrology, Calibration Chain of Custody & Statutory Survey Report Sign-Off

NTA reference: GN 156

Calibration chains, sensor fault-finding, poisoned cells, report authorship and sign-off.

The A.I.A. capstone is metrology governance: calibration chain of custody, fault-finding on poisoned sensors, and authoring a statutory-style survey report that a client and an inspector can both rely on.

Practical video recording scheduled — the online reader reserves a demonstration slot for this module.

ASSESSMENT

Assessment framework

Instrument practicals with calibration records, exposure calculation papers, and a signed statutory-style survey report portfolio.

L6-M1	Module paper + portfolio evidence	Equal
L6-M2	Module paper + portfolio evidence	Equal
L6-M3	Module paper + portfolio evidence	Equal
L6-M4	Module paper + portfolio evidence	Equal
L6-M5	Module paper + portfolio evidence	Equal
L6-M6	Module paper + portfolio evidence	Equal
L6-M7	Module paper + portfolio evidence	Equal
L6-M8	Module paper + portfolio evidence	Equal

Certification: Attendance and Completion certificates are issued internally; Accredited certification follows only after NQA status is granted for this qualification.